Ransomware Incident Analysis

Course: Google Cybersecurity Professional Certificate
Project Type: Incident Handler's Journal
Scenario: Simulated Ransomware Attack on a Healthcare Clinic

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1. Project Overview

This project involved analyzing a simulated ransomware incident affecting a small healthcare clinic in the United States.

The objective was to document the incident, identify how the attackers gained access, and understand the impact on the organization's systems and business operations.

2. Incident Handler's Journal

Journal Entry: 1
Date Recorded: September 5, 2026
Incident Type: Ransomware Attack
Organization: Small U.S. Healthcare Clinic (Simulated)

Incident Description

The incident involved a ransomware attack against a small healthcare clinic.

Employees were unable to access medical records and other important files because attackers had encrypted the organization's data.

The attackers gained access through targeted phishing emails sent to several employees. These emails contained malicious attachments that installed malware when downloaded.

After gaining access, the attackers deployed ransomware, encrypted critical files, and demanded payment in exchange for a decryption key.

The incident disrupted the clinic's operations and prevented employees from accessing essential systems.

Tools Used

None. This activity involved reviewing a provided incident scenario and documenting the findings. No security investigation tools were used.

3. Incident Analysis — The 5 W's

Question	Findings
Who?	An organized group of malicious attackers known to target healthcare and transportation organizations.
What?	A ransomware attack encrypted important files and prevented employees from accessing medical records and other critical information.
When?	Tuesday morning, approximately 9:00 AM.
Where?	A small healthcare clinic in the United States.
Why?	Attackers sought financial payment by encrypting critical files and demanding a ransom for the decryption key.

4. Attack Sequence

Based on the provided incident scenario, the attack followed this sequence:

1. Attackers sent targeted phishing emails to employees.
2. The emails contained malicious attachments.
3. Employees downloaded the attachments, installing malware.
4. Attackers gained access to the organization's network.
5. Ransomware was deployed and encrypted critical files.
6. Employees lost access to important systems and medical records.
7. The attackers demanded payment in exchange for a decryption key.

5. Additional Notes and Recommendations

My main recommendation was to provide employees with **security awareness training**.

This would help employees recognize suspicious emails, identify potentially malicious attachments, and reduce the likelihood of successful phishing attacks.

The incident demonstrated how phishing can lead to ransomware infections and disrupt essential healthcare services.

6. Skills Demonstrated

- Incident Documentation
- Ransomware Analysis
- Phishing Attack Identification
- Incident Handler's Journal
- The 5 W's Framework
- Security Awareness
- Incident Response Fundamentals

7. Conclusion

This project provided experience documenting and analyzing a simulated cybersecurity incident.

I learned how phishing attacks can lead to ransomware infections, how attackers can disrupt business operations, and why accurate incident documentation is important.

The exercise also reinforced the importance of employee security awareness in reducing phishing-related security risks.

This project was completed as part of the Google Cybersecurity Professional Certificate using a fictional incident scenario. The analysis is based on my original incident handler's journal entry.